

Syllabus: Acoustics, Electrical, and Illumination Systems
Catalog Number: ARC 5443
Instructor: Kurt Neiswender, AIA
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Office Hours: By Appointment
Telephone:
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Course Meetings:

Lecture/Lab: Thursdays: 7:30 – 11:35 PM
Room: Zoom, See Canvas for links

Canvas: Online course management

The course syllabus, course calendar, assignments, additional readings and images, and other notes will be posted on the course Canvas site. Students are responsible for monitoring the site on a regular basis and for keeping up with coursework. You are expected to have your camera on 100% of the class session to receive attendance and participation credit for this course.

Catalog Description

The course reviews the theories of sound perception, analysis of architectural acoustics, room acoustics, sound isolation, sound absorption, electronic sound reinforcement, and the study of acoustic properties of materials and room shapes to provide speech privacy, enhance speech clarity, enrich music spaces and control noise. The course also addresses basic electrical theory, utility connection options, emergency backup, distribution, and safety considerations as they relate to building construction. Electrical lighting is studied as part of an assimilated section covering the fundamentals of illumination, integrating natural daylighting and electrical lighting design for energy conservation. Class projects include investigations into course subjects to further enhance understanding of real work applications.

Prerequisites

Architecture Foundations Studio 2 - ARC 5024, Architecture Interiors Studio 1 - ARC 5113

Course Content

This course is part of the building systems series and covers topics of lighting, electricity and acoustics systems integration in building design. The fundamental scientific principles governing light and sound in the built environment will be examined in the context of human physiological, psychological and biological needs as well as sustainable design perspective. It exposes students to technologies, materials and strategies for control of light and sound in buildings as well as the basic analyses needed to inform design decision-making and examine project performance. The course will continue the themes of resource efficiency and end-user comfort through the examination of emerging metrics for daylight sufficiency, visual, and acoustic comfort. The content is divided into key topics, progressing logically from foundational concepts to advanced systems, with practical applications and case studies. The knowledge gained in the systems courses are integral to your success in your Comprehensive Design Studio.

Required Text(s)

Mechanical & Electrical Equipment for Buildings (MEEB) **13th Edition**
Walter T. Grondzik and Alison G. Kwok
John Wiley and Sons, Inc. 2019

Recommended Text(s) and Resources

Designing with Light: The Art, Science and Practice of Architectural Lighting Design,
Jason Livingston
Carbon-neutral Architectural Design 2nd Edition, Pablo La Roche
The Coffee Sketch Podcast – www.coffeesketchpodcast.com

NAAB Program Criteria (PC)

Technical Knowledge: How the program ensures that students understand the established and emerging systems, technologies, and assemblies of building construction, and the methods and criteria architects use to assess those technologies against the design, economics, and performance objectives of projects.

Environmental Systems: (SC.4B)

Principles of environmental system design: Student is able to demonstrate understanding of basic principles of environmental systems design, based on qualitative and quantitative reasoning (I 01)

Application to problems: Student is able to demonstrate understanding of application of environmental systems principles and technologies to problems involving solar geometry, daylight, natural ventilation, heating and cooling, indoor air quality, lighting systems, and acoustics (R 02)

Geographic regions: Student is able to demonstrate understanding of using analysis to address environmental systems conditions across different geographic regions (E 03)

Building Service Systems: (SC.4E)

Principles of building service systems: Student is able to demonstrate understanding of basic principles of building service systems (including mechanical, plumbing, fire protection, electrical, and lighting systems) (I 01)

Application of building service systems: Student is able to demonstrate understanding of the appropriate application of building service systems (R 02)

Analysis of building service systems: Student is able to demonstrate understanding of performance analysis of building service systems through quantitative reasoning (R 03)

Application to problems: Student is able to demonstrate understanding of application of building service systems to practical and theoretical problems using advanced technologies (E 04)

Course Objectives

The course work during this semester will emphasize building systems integration:

1. **Environmental Systems:** Ability to demonstrate the principles of environmental systems' design, how design criteria can vary by geographic region, and the tools used for performance assessment. This demonstration must include active and passive heating and

- cooling, solar geometry, daylighting, natural ventilation, indoor air quality, solar systems, lighting systems, and acoustics. (SC.4B)
2. **Building Service Systems:** Understanding of the basic principles and appropriate application and performance of building service systems, including lighting, mechanical, plumbing, electrical, communication, vertical transportation, security, and fire protection systems. (SC.4E)

Additional Learning Outcomes

- **Ecological Knowledge:** Understanding of the dynamic between built and natural environments, of the impacts and the mitigation of climate change, and of the principles of resilience design.
- **Site Design:** Ability to respond to site characteristics, including urban context and developmental patterning, historical fabric, soil, topography, ecology, climate, and building orientation, in the development of a project design.
- **Health, Safety, and Welfare in the Built Environment:** How the program ensures that students understand the impact of the built environment on human health, safety, and welfare at multiple scales, from buildings to cities.
- **Building Integration:** How the program ensures that students develop the ability to make design decisions within architectural projects while demonstrating integration of building envelope systems and assemblies, structural systems, environmental control systems, life safety systems, and the measurable outcomes of building performance.

The course will be conducted in both lecture/seminar and lab formats which will involve theory, practical applications, calculations, and appropriate hands-on experiments with performance modeling tools. Topics are explained in lectures, practiced in the labs and integrated in the design process with the included assignments.

Common Course Assignments and Projects

Course assignments are outlined below and will be posted to Canvas throughout the semester, and will be posted on Canvas. Each assignment is due on the designated day and time. Assignments will be listed on Canvas with the final due date and time. All completed assignments will be submitted digitally via Canvas. Do not email assignments as attachments. Due dates are tentative and may be adjusted in order to allow for coordination with students' studio work. Late assignment will be marked down 2.5% per day late. All assignments shall be graded for quality and content based on rubrics attached in Canvas. Sloppy, illegible, and disorganized works are not acceptable for assignment and shall negatively impact your course grade.

Assignments:

1. **Assignment #1:** Cove Introduction and basic input/output analysis. Daylighting Analysis
2. **Assignment #2:** Acoustical Design Spatial Configuration Project

Evaluation of Student Work

Evaluation rubric will be posted on Canvas for each assignment of the project. Quizzes and exams will be automatically/manually evaluated and the results will be posted on Canvas. Quizzes and exam will be multiple choice, true or false, fill in the blank, and will include calculations and plots. Two tests will be given, midterm and final exam will be taken during the university assigned final exam week. No make-up test or exam, especially the final exam must be taken to pass the course.

Course Software

Cove: www.cove.tools

"At Cove, we provide expert architectural and sustainability consulting that's both cost-effective and efficient. Our AI-powered methods and advanced analytics accelerate projects without sacrificing quality. Trust us to deliver superior value and exceed your expectations with every project".

To access LTU's licenses for Cove please go to the website above and create a new user account using your @ltu.edu email account. It should connect you to our account. If it does not please let me know as soon as possible. Many of you are already signed up, no new account work necessary.

Rhino/Grasshopper: pachyderm plug-in
Revit: Lighting and Electrical Design

Course Grading Policy

The grading policy is as follows:

Attendance:	10%
Discussion Participation:	10%
Quizzes:	20%
Assignments:	30%
Exams:	30%
	100%

General grading rules:

A	94% - 100%
A-	< 94% - 90%
B+	< 90% - 87%
B	< 87% - 84%
B-	< 84% - 80%
C+	< 80% - 77%
C	< 77% - 74%
C-	< 74% - 70%
D+	< 70% - 67%
D	> 67% - 64%
D-	< 64% - 61%
F	< 61%

Exception:

The instructor reserves the right to adjust the low-end average cut-off score based upon the statistical distribution of the semester averages for final grades. For instance, if there is a large gap in the final averages at 91%, then the instructor may decide to award an A grade to all students above that value. This can be done at the instructor's discretion and in no way should be construed to mean that it will be done each and every semester.

Attendance:

Students can miss a maximum of 20% of a course. This equates to 9 absences in a class that meets 3 times a week; 6 absences in a class that meets 2 times a week; 3 absences in a class that meets 1 time a week. The absences are counted separately for multi-component courses. For example, a student can miss 3 of 15 lab sessions and 6 of 30 studio sessions. If a student exceeds these absences, they will fail the course and receive a grade of F. For additional detail refer to the Student Companion linked below.

Instructor will notify you at the first and second absence. Once a student reaches a third absence you will be reported to the registrar's office and given a failing grade for non-attendance. Your instructor will review understanding of the syllabus in the first class session for mutual understanding and clarification.

The Student Companion

Student Conduct, Academic Honesty, and Department Policies

All of the college's policies and procedures have been compiled into a single document, The Student Companion. All students are responsible for reading and understanding these policies. The Companion is posted as a downloadable pdf document online for all faculty, staff, and students:

https://www.ltu.edu/architecture_and_design/student_resources.asp

Topics covered in the Student Companion include:

- The Studio Code
- Plagiarism
- Attendance and Late Work
- Participation

Graduate Online Courses: Attendance, Participation, and Engagement [Required for Graduate online courses]

CoAD Graduate Courses are comprised of asynchronous and synchronous educational engagement structures so that knowledge can be explored in a variety of ways. In order to maintain the high, rigorous standards of the CoAD academic environment, consistent attendance and participation in both synchronous and asynchronous modalities is a necessity. Required synchronous class sessions will be held within the class times listed on Banner, while other class meetings may be arranged based on student-faculty agreement. In the case that conflicts occur between a listed course time, synchronous content delivery, and extra-curricular student responsibilities, the student should communicate directly with their faculty member. The student is responsible to compensate for these scheduling conflicts during the semester. Since graduate work is assessed on performance, a lack of engagement through absence will inevitably impact the educational outcomes.

Course Schedule

Week	Part	Key Topics	Asynchronous Activities (Before Zoom)	Synchronous Zoom Activities (Live Session)	Major Due Dates / Assignment Status
1	Foundations & Daylighting	Course overview, NAAB criteria; Design process & systems integration; Solar geometry (sun paths, angles, site analysis); Intro to daylighting (principles, benefits, metrics); Intro to Cove.tool.	Recorded Lectures; Reading: MEEB Ch 1, Ch 10; Syllabus review; Quiz 1.	Icebreaker; Discussion on site/orientation & daylight; Lucidspark: Sun path diagrams; Cove.tool demo.	Homework Week 1: Daylight and Passive Design Analysis
2	Foundations & Daylighting	Advanced daylighting metrics; Daylighting strategies (sidelighting, toplighting, light shelves, shading); Cove.tool for iterative design; Qualitative aspects of daylight.	Recorded Lecture; Reading: MEEB Ch 10, Livingston (daylight); Quiz 2 (covers MEEB Ch 1 & 10).	Q&A on Cove.tool/metrics; Lucidchart: Diagramming daylighting strategies; Cove.tool Workshop. Discussion: Balancing daylight, glare, thermal gains.	Homework Week 2: Daylight Analysis in Cove Tool
3	Illumination Systems	Physics of light, human eye/vision; Lighting terminology (lumen, lux, CRI, CCT); Lighting levels/quality; Electric light sources (incandescent, fluorescent, HID, LED).	Recorded Lectures; Reading: MEEB Ch 6, Ch 15; Livingston (Part 1); Quiz 3 (covers MEEB Ch 6).	Discussion on light source technologies; Lucidspark: Matching light sources to goals; Virtual "Show and Tell" of lighting examples.	No Homework

4	Illumination Systems	Luminaire types & characteristics; Photometric data (IES files, candela curves); Basic lighting calculations (Lumen Method/Zonal Cavity); Intro to lighting design software.	Recorded Lectures; Reading: MEEB Ch 16; Livingston (Part 2); Quiz 4 (covers MEEB Ch 15 & Photometrics).	Q&A on photometrics; Lucidchart: Sketching luminaire layouts; Hands-on: Lumen Method calculation; Revit Demo (optional).	Homework Week 4: Lighting System Analysis
5	Illumination Systems	Lighting design process (conceptualization, SD, DD); Layering light (ambient, task, accent); Integrating daylight & electric light; Lighting controls; Case studies.	Recorded Lectures; Reading: MEEB Ch 17; Livingston (Part 3); Quiz 5 (covers MEEB Ch 16).	Lucidspark: Brainstorming lighting layers; Discussion on lighting & architectural expression.	Homework Week 5: Lighting Design for Museum Plan
6	Electrical Systems	Basic electrical theory (V, I, R, P, Ohm's Law, Watt's Law); AC/DC; Circuits; Electrical safety; Overview of building electrical systems (service, distribution, branch circuits); Components.	Recorded Lectures; Reading: MEEB Ch 25, Ch 26 (overview); Quiz 6 (covers MEEB Ch 17 & Basic Electrical Principles).	Q&A on electrical concepts; Lucidchart: Diagramming series/parallel circuits; Discussion on electrical hazards/safety; Virtual tour of electrical room/panel.	Homework #6: Panel Schedule Exercise Assignment #2 Presentation
7	Electrical Systems	Electrical load calculations (NEC, demand factors); Sizing conductors & overcurrent protection; Branch circuit design (lighting, receptacles); Wiring diagrams & symbols; Intro to Revit electrical.	Recorded Lectures; Reading: MEEB Ch 28; Quiz 7 (covers MEEB Ch 25 & 26).	Workshop: Branch circuit design; Lucidchart: Electrical one-line diagram & wiring layout; Revit Demo (optional). Assignment #1 Review	Homework #7 Riser Diagramming
8	Architectural Acoustics	Physics of sound (wavelength, frequency, dB); Sound perception & psychoacoustics; Sound reflection, absorption, diffusion, diffraction; Reverberation time (RT60); Intro to sound level meters.	Recorded Lectures; Reading: MEEB Ch 7; Quiz 8 (covers MEEB Ch 28).	Discussion on good/bad acoustic spaces; Lucidspark: Identifying acoustic challenges; Demo: Intro to sound meter exercise.	Homework #8: Acoustic Sound Samples with Decibel X Phone App
9	Architectural Acoustics	Room acoustics (speech, music, privacy); Acoustic properties of materials (NRC, STC); Sound absorption & isolation strategies; Building noise control (HVAC, plumbing, impact); Intro to Pachyderm.	Recorded Lectures; Reading: MEEB Ch 22, Ch 23; Quiz 9 (covers MEEB Ch 7).	Case Study Analysis: Acoustical design; Lucidchart: Diagramming sound paths & noise control; Pachyderm Demo (optional).	Homework #9: Acoustics analysis Custom Application/Pachyderm
10	Integration & Review	Coordination of lighting, electrical, acoustic systems; Space planning for systems; Comprehensive design goals; Review of major concepts; Final project workshop.	Recorded Lectures; Reading: MEEB Ch 29; Review all course materials.	Open Q&A/Review; Lucidspark: "Key Takeaways & Connections"; Breakout Rooms: Peer review of Assignment #4; Instructor consultations.	Homework #10: Noise Control Applications